Where the Flood Reached: Terrain-Only Classical Machine Learning of Building Damage in the 2026 Langtang Lirung–Trishuli Debris Flood, Nepal

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***Abstract*—On 26 August 2026 a rock–ice avalanche from Langtang Lirung became a debris flood that ran more than 90 km down the Bhote Koshi–Trishuli valley, Nepal. We ask how well terrain alone, from a 30 m elevation model, ranks buildings by recorded damage and transfers to an unseen part of the valley. Copernicus EMSR927 damage grades were joined to pre-event Overture footprints: 17,503 building records (3,360 affected, 2,505 destroyed) in three reaches, each in turn held out. Over all buildings, class-weighted logistic regression and a linear SVM transferred best (held-out ROC-AUC 0.96–0.98 for affected), but much of that score comes from buildings far above the local damaging height. Among near-channel buildings (within 300 m of the flood path, below roughly the local damaging height) terrain still ranks affected buildings in an unseen reach with AUC 0.85–0.88 in the upper and middle reaches (0.80 in the lower, falling at tighter limits) and separates destroyed from damaged buildings with 0.80–0.93 (0.72 in the lower reach, whose destroyed labels are doubtful); there, height above the flood path (HAND) alone is not detectably worse than the full model. The fitted attenuation curve implies a damaging height falling by 19.5 m (affected) and 23 m (destroyed) per doubling of along-river distance (intervals overlapping), with no data between 16 and 42 km. The ranking transferred but the probability level did not: the unweighted logistic regression under-predicted upstream and over-predicted downstream, and re-fitting the intercept alone on 20–50 labelled buildings spread over the reach typically removed most of this error for both the unweighted and the class-weighted model. Terrain gives a transferable ranking; the probability level must come from the event.**

***Index Terms*—debris flood, building damage, Height Above Nearest Drainage, logistic regression, spatial transfer, calibration, area of applicability, Nepal Himalaya.**

I. Introduction

At 08:37 local time on 26 August 2026, a rock–ice mass on the north face of Langtang Lirung failed at about 5,100–5,200 m, fell into the Lhende Khola valley and became a debris flood along the Bhote Koshi and Trishuli rivers [1]. Within an hour it destroyed the Rasuwagadhi border crossing, damaged several hydropower plants and struck settlements from Timure to Betrawati and beyond [1, 2]. Events of this kind recur across High Mountain Asia: one inventory lists 60 large rock and ice avalanches, 29 of which cascaded into floods or debris flows [3], and the 2023 South Lhonak outburst in Sikkim showed how a single high-mountain failure can damage infrastructure a hundred kilometres downstream [4].

a) What is known: Studies of Himalayan cascades reconstruct the physical event and count what was exposed. They overlay modelled or mapped flood extents on buildings, roads and hydropower [2, 4–6], or survey losses after the event [7]. Building-level damage models exist for floods and debris flows elsewhere, but they rely on simulated or surveyed flow depth, velocity and impact pressure [8–11]. In the studies reviewed here, machine-learning damage models for Nepal exist only for earthquakes, where structural survey data are available [12]. Nepal flood-susceptibility studies use related factors but map land; they do not model buildings [13]. Table I compares these studies.

b) How they are tested: Three weaknesses recur. Many models are fitted and scored on the same buildings: the vulnerability curves of Wang et al. [11] exceed 85% accuracy on the 36 buildings they were fitted to, and Barnhart et al. [10] test on the single event that motivated them. Where machine learning is used, train and test buildings are often drawn at random from the same area; Currie and Reinke [14] show that skill falls from a median macro-F1 of 0.68 to 0.54 once whole events are held out. And damage is rare, so accuracy on imbalanced data can look good while missing most damaged buildings [15, 16].

c) Why terrain: Terrain is the only input known everywhere and in advance, at the scale of single buildings. Height Above Nearest Drainage turns an elevation model into the height a flood must rise to reach each point, and underlies national flood-inundation mapping in the United States [17]. In a steep valley, where a debris flood is confined between valley walls, that height should matter even more than on a lowland floodplain; how well it predicts actual building damage, and how the height at which damage becomes likely changes downstream, is not known.

d) The gap: Four gaps follow from Table I. (i) None of the Himalayan cascade studies reviewed here learns from *observed* building damage; they count exposure, and none predicts damage. (ii) Damage models need hydraulic intensities that do not exist for a fast, unmonitored cascade; whether terrain-only proxies such as Height Above Nearest Drainage (HAND) [17] can stand in is untested. (iii) Damage models lose accuracy when moved to new places [8, 14], and Ghimire et al. [18] name this host-to-target transfer as an open problem.

Along a single cascade the flood weakens downstream, so the loss should be systematic and measurable. (iv) Event size controls how many buildings are damaged [10], yet a terrain-only model knows nothing about event size. How such a model should be generalised has not been examined.

e) Aim and research questions: We build and evaluate terrain-only models of observed building damage for the 2026 cascade with transparent, classical machine learning, and measure how they transfer along the valley. The questions are:

RQ1. How well do pre-event terrain features predict whether a building was *affected*, whether it was *destroyed*, and, among affected buildings, whether it was destroyed rather than damaged?

RQ2. How does the height above the flood path at which damage becomes likely (HAND_{50}) change with distance downstream?

RQ3. Which classical learners transfer best to an unseen reach, and why?

RQ4. What transfers to a new reach, or a new flood: the ranking of buildings, the probabilities, or both?

f) Contributions: (1) A terrain-only, classical machine-learning model of recorded building damage for a Himalayan rock–ice debris-flood cascade, evaluated on an unseen reach both over all buildings and, as the fairer measure, over the near-channel buildings, where the dominance of hillside negatives is reduced. (2) A quantified attenuation curve of the damaging height above the flood path. (3) A transfer test along one cascade, with a partial explanation of why flexible models fail. (4) A separation of what transfers (the ranking) from what does not (the probability level, and with it the decision threshold), with a cheap way to recalibrate.

g) Definitions: Affected means any Copernicus grade (destroyed, damaged or possibly damaged). *Reach* is one of three river sections (upper, middle, lower; Section II). *Transfer* means training on two reaches and testing on the third. *Flood path* is the steepest-descent flow path (D8, eight-direction routing) traced on the elevation model from the border crossing at Rasuwagadhi (Section III). *HAND* is the height of a building above the cell of the flood path that it drains to; the reference is the flood path, although the name says nearest drainage. *Near-channel* buildings are those within 300 m of the flood path and below a reach-specific HAND limit (Section III). Three quantities could each be called a threshold, so they are named separately throughout: the *damaging height* HAND_{50} is the height above the flood path at which half the buildings were affected (Section V-G); the *decision threshold* is the probability above which a building is flagged; and the *probability level* (calibration-in-the-large) is how high the predicted probabilities sit as a whole.

Section II describes the study area and data, Section III the methods, Section IV the experimental design, Section V the results, Section VI the discussion and Section VII the conclusions.

II. Study area and data

A. Study area

The study covers the Bhote Koshi–Trishuli corridor in Rasuwa and Nuwakot districts, from the border at Rasuwagadhi to the end of the Phosretar mapping area, 91 km downstream along the flood path (Fig. 1). The upper part is a steep, narrow gorge; below Betrawati the valley widens and the channel flattens. The four Copernicus mapping areas form three reaches (Table II) and leave a gap between about 16 and 42 km, where no analysed buildings lie.

The reaches differ in setting as much as in distance; the ranges below are interquartile ranges over buildings. The upper reach (Syaprubesi and Timure) is a gorge: the channel falls 2.2–2.9% at the buildings and the valley floor (HAND $\leq$ 20 m) is 130–155 m wide, with settlements on terraces and fans. In the middle reach (Betrawati–Bidur) the slope drops to 0.5–1.0% and the floor widens to 190–260 m, with dense settlement close to the channel. The lower reach (Phosretar) is flatter (0.2–0.4%) and wider (210–390 m). The share of buildings affected falls from 77% upstream to 9% downstream (Table II).

The event itself was poorly observed: there were no instruments above 5,500 m at the source, and warnings reached Timure only after it had flooded [1]. No flow depths, velocities or deposit thicknesses for the settlements were available to this study, so the hydraulic inputs of physically based damage models [10, 11] cannot be used.

B. Data

All data are open or openly licensed for research (Table III). The study uses no personal data: buildings are anonymous footprints, damage grades come from satellite interpretation, and no people were surveyed, so no ethical approval was needed.

Pre-event footprints came from the Overture Maps release of 19 August 2026, one week before the event, which merges OpenStreetMap with machine-detected footprints; post-event mapping would miss destroyed buildings. Each footprint is reduced to its centroid, where every feature is read. Footprint shape, size and material were deliberately left out: the model sees only where a building stands, which is all that is known consistently for a valley not yet hit, and size is unknown for the point-only rows of the label join below.

a) Labels: The Copernicus grading products list affected buildings only. Each graded point was matched to the footprint containing it, or to one within 10 m, keeping the worst grade per footprint; 3,400 of 3,816 points (89%) matched, on 2,944 distinct footprints. The 416 unmatched points, mostly buildings absent from the pre-event footprints, are kept as point-only rows. Footprints with no graded point are labelled *not affected*. Table IV maps the grades to the three targets.

b) Cleaning: Three rules remove ungraded footprints only (18,908 $\rightarrow$ 17,503 rows; Table XVII): R1 removes 1,282 inside the Copernicus flood trace, whose status is ambiguous; R2 removes 106 smaller than 10 m^2; R3 removes 17 within 8 m of an affected footprint from another source (duplicates). Because R1 uses post-event information to select rows, the

TABLE I
COMPARISON OF RELATED STUDIES. PERFORMANCE IS GIVEN WHERE THE PAPER REPORTS A SINGLE FIGURE; "–" MEANS THE STUDY IS DESCRIPTIVE OR REPORTS NO COMPARABLE FIGURE. LR: LOGISTIC REGRESSION; SVR: SUPPORT VECTOR REGRESSION; RF: RANDOM FOREST; GB: GRADIENT BOOSTING; ANN: ARTIFICIAL NEURAL NETWORK; TWI: TOPOGRAPHIC WETNESS INDEX; SPI: STREAM POWER INDEX; HMA: HIGH MOUNTAIN ASIA.

Study	Hazard / region	Feature extraction	Model	Performance	Limitations
Paulik et al. [8]	Floods (6 events)	Up to 7 hazard and building variables, observed depth	Random forest regression; spatial transfer test	Precision falls when transferred between flood types	Needs observed depth; building attributes
Wüthrich et al. [9]	Flash flood, Germany	Field-surveyed depth, velocity, distance to bank, construction	Statistical analysis	–	Not predictive
Barnhart et al. [10]	Debris flow, USA	Simulated depth, momentum flux (3 runout models)	Fragility functions	Best forecast: hit rate 48%, threat score 25%	Needs runout model and event volume; one event, no transfer test
Wang et al. [11]	Debris flow, SW China	FLO-2D depth, velocity, impact pressure; field survey	Lognormal vulnerability curves (vs. 4 other forms)	Prediction accuracy >85% (in-sample fit)	36 damaged buildings, one village; no held-out test
Ghimire et al. [12]	Earthquake, Nepal	Structural parameters + ShakeMap intensity	LR, SVR, RF, GB	68% accuracy (3 classes); 64% with 4 features	Needs structural survey
Ghimire et al. [18]	Earthquake, Italy	Age, storeys, area, height + intensity	RF, GB, XGBoost	72% (all) / 68% (basic) accuracy	Needs structural data; transfer untested
Currie and Reinke [14]	11 disasters, global; floods excluded	Sentinel-1 coherence, intensity and texture	Random forest; leave-one-event-out	Median macro-F1 68% → 54% on unseen events	Needs post-event imagery; uncalibrated scores
Duwal et al. [13]	Riverine flood, Nepal	10 conditioning factors incl. TWI, SPI, distance to river	SVM, RF, ANN	SVM best by AUROC	Land, not buildings
Sattar et al. [4]	Outburst cascade, India	Remote sensing, hydrodynamic modelling	Physical reconstruction	–	Impacts counted, not predicted
Zhong et al. [3]	Rock–ice avalanches, HMA	Event inventory (60 events)	Inventory statistics	–	Event level; no building impacts
Gouli et al. [5]	Outburst floods, China–Nepal	Lake and terrain factors	ML susceptibility + hydrodynamic model	–	Exposure, not damage
Chen et al. [6]	Outburst floods, Nepal	Multi-temporal imagery	RF, Bayesian regression, flood model	–	Exposure only
Li [2]	**This event**	Precipitation, temperature, UNOSAT damage	Exposure overlay	–	No predictive model
This study	**This event**	10 terrain features from a 30 m elevation model (9 in the attenuation set)	9 classical learners; leave-one-reach-out	ROC-AUC 0.96–0.98 (affected, unseen reach); 0.85–0.88 near the channel in the upper and middle reaches, 0.80 in the lower	Single event; photo-interpreted labels; part of the skill is flood extent

TABLE II
THE THREE REACHES USED FOR TRANSFER TESTS. DISTANCE IS ALONG THE FLOOD PATH FROM THE BORDER.

Reach	km	Bldgs	Affected	Destr.	Blocks
Upper[a]	0–16	845	653 (77%)	531	12
Middle[b]	42–65	10,456	2,141 (20%)	1,699	38
Lower[c]	65–91	6,202	566 (9%)	275	40
Total		17,503	3,360 (19%)	2,505	90

[a]Syaprubesi and Timure; [b]Betrawati–Bidur; [c]Phosretar. The middle and lower areas meet at km 65; their boundary is not perpendicular to the flood path, so they overlap by about 0.4 km. Blocks: 1-km spatial blocks.

results with the R1 rows included are reported beside the main ones (Table VIII, Section V-K).

c) Label audit: A stratified sample of 200 buildings (50 per area; 80 destroyed, 40 damaged or possibly damaged, 80 not affected) was checked against pre- and post-event Vantor imagery at 0.2 m/pixel; 41 chips could not be judged, mostly because of cloud, missing post-event images or no visible building. Of the 80 "not affected" footprints, 15 showed no building on either date (10 of 40 in the upper reach, 1 of 20 in the middle, 4 of 20 in the lower), so some of the table's negatives are not buildings; all 62 judgeable ones appeared intact. "Destroyed" agreed in 52 of 52 judgeable cases in the upper and middle reaches, almost always as a footprint under a uniform sediment or boulder field. In the lower reach, 8 of 18 judgeable "destroyed" buildings still stood, often in a row of standing houses surrounded by sediment, so the *destroyed* target is least certain there. Of 27 judgeable "damaged" or "possibly damaged" buildings, 7 showed visible damage and 20 looked intact from above, which a roof-level view cannot rule out.

d) Building counts: Our building table and the Copernicus inventory do not count the same buildings (Table V). Prevalence agrees in the middle reach (20.5% in both, from 10,456 and 11,407 records); in the lower reach our table holds 6,202 rows against 4,502 inventory buildings, mostly extra machine-detected footprints. Lower-reach prevalence is therefore 9.1% in our table, about 11% without the audit's share of empty footprints, and 13.5% in the inventory. The 17,503 rows are building records and have not been verified as buildings.

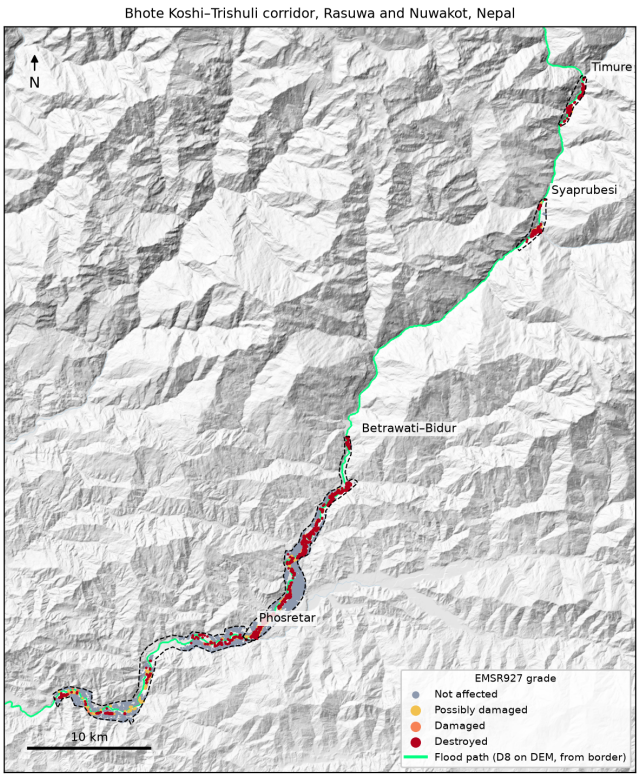


Fig. 1. Study area. Dashed outlines are the four Copernicus EMSR927 mapping areas. Points are pre-event buildings coloured by damage grade; the green line is the flood path, the D8 flow path traced on the elevation model from the border crossing (not the Copernicus flood trace).

e) Elevation model: GLO-30 is a surface model and includes canopy; FABDEM, which removes forests and buildings, gives HAND 1.5 m lower at the median but changes model AUCs by less than 0.002 (Section V-K), so GLO-30 is kept.

III. Methods

Fig. 2 summarises the pipeline. Three inputs (damage grades, footprints and the elevation model) are reduced to one building table with ten terrain features and three binary targets, and land cover enters only the error analysis; every experiment of Section IV reads that table. Most are scored under the leave-one-reach-out protocol described below; the attenuation curve, the regime analysis and the descriptive interpretation fits use all buildings.

A. Terrain features

The GLO-30 tiles were reprojected to UTM zone 45N at 30 m, pits and flats were resolved, and D8 flow directions were computed with pysheds. The flood path is derived from the elevation model alone: it is the D8 flow path traced from one seed cell, the main channel at the Rasuwagadhi border crossing, downstream to the edge of the grid, plus 15 km up the main stem beyond the border, so that the few buildings just upstream of the seed also have a path cell (their along-river distance is negative). It uses no post-event information beyond the choice of seed, that is, which river the flood came down, and it is distinct from the Copernicus flood trace, which is used only for cleaning rule R1. The path therefore leaves out tributaries such as the Tadi Khola, and HAND, distance and along-river distance are measured to it. A definition with no seed at all, measuring to every river above a drainage-area threshold, is tested in Section V-K. Ten features form the *main* set (Table VI); stream power was dropped ($\rho = 0.99$ with channel slope). An *attenuation* set replaces the valley-geometry group with $\log(d + 10\,\mathrm{km})$, where d is distance along the river, and its product with HAND. Because the reaches lie in order along the river, $\log(d + 10\,\mathrm{km})$ is extrapolated whenever the

TABLE III
Data sources.

Data	Source	Role	Licence
Damage labels	Copernicus Emergency Management Service (EMS) activation EMSR927, areas of interest (AOIs) 01–03 and 05[a] [19]	3,816 points graded destroyed / damaged / possibly damaged by photo-interpretation	Copernicus EMS terms (free and open)
Footprints	Overture Maps 2026-08-19 [20]	Pre-event buildings; used only to place labels and define undamaged buildings	ODbL 1.0
Elevation	Copernicus GLO-30 [21]; FABDEM [22]	Source of all features; FABDEM as robustness check	Copernicus DEM licence (free); CC BY-NC-SA 4.0
Imagery	Vantor Open Data (OpenAerialMap), 2021–2026	Visual label audit only	CC BY-NC 4.0
Land cover	FRTC/ICIMOD NLCMS 2022 [23]	Error analysis only	CC BY 4.0
Land cover (check)	ESA WorldCover 2021 v200 [24]	Stand-in for NLCMS in the error analysis (Appendix G)	CC BY 4.0

[a]The activation has six AOIs; only these four have a building grading product, so AOIs 04 and 06 provide no building labels.

TABLE IV
Copernicus damage grades and the three binary targets (counts after cleaning).

EMS grade	n	Affected	Destroyed	Destr. | aff.
Destroyed	2,505	1	1	1
Damaged	282	1	0	0
Possibly damaged	573	1	0	0
No graded point	14,143	0	0	–

TABLE V
Building counts and prevalence of *affected* under three denominators: our table; our table less the audit's share of empty footprints (95% Wilson range); the Copernicus inventory (affected counts are graded records).

	Our table		Audit-	Copernicus	
Reach	n	Prev.	corrected	n	Prev.
Upper	845	77.3%	81.9% (80–85)	1,000	86.4%
Middle	10,456	20.5%	21.3% (21–25)	11,407	20.5%
Lower	6,202	9.1%	11.2% (10–15)	4,502	13.5%

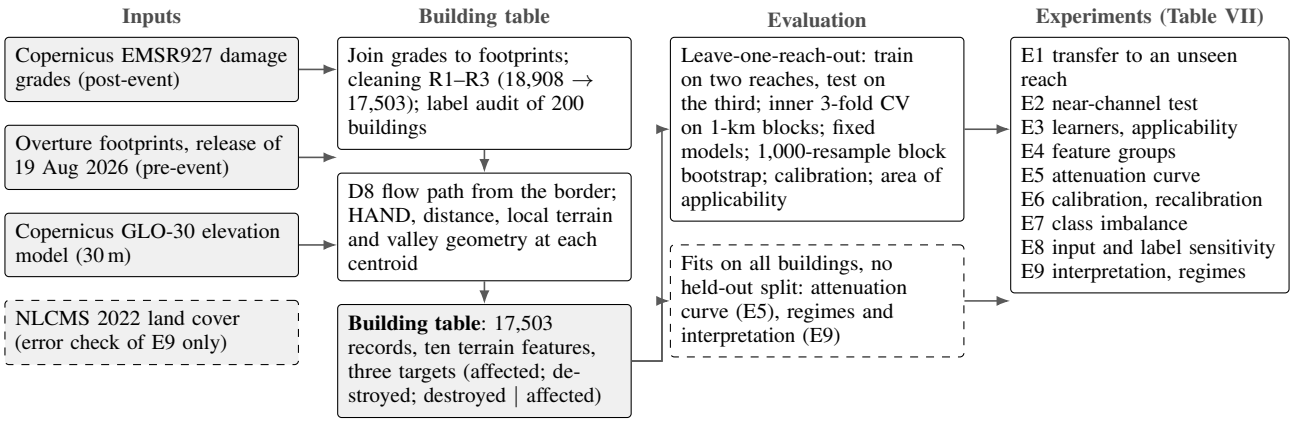


Fig. 2. Workflow. The first three rows of Table III (damage grades, footprints and the elevation model) are joined into one building table of 17,503 records with ten terrain features and three binary targets; land cover enters only the error analysis. One reach at a time is held out, with tuning inside 1-km spatial blocks; the experiments E1–E9 are listed in Table VII.

TABLE VI
TERRAIN FEATURES. ALL COME FROM THE ELEVATION MODEL.

Group	Features	Physical rationale
Core	HAND; distance to flood path	Vertical and horizontal exposure
Local	Slope; relief; profile and plan curvature; topographic wetness index	Terraces and fans vs. hillsides; convex fans and spurs vs. hollows that concentrate flow
Geometry	Valley width (HAND ≤ 20 m, log); channel slope; signed bend exposure	Confined gorges give deeper, faster flow; outer banks erode
Attenuation[a]	$\log(d+10)$; HAND $\times \log(d+10)$	The flood weakens downstream

[a]Replaces the geometry group in the attenuation set.

upper or the lower reach is held out.

HAND is the elevation of a cell minus that of the flood-path cell it drains to along D8 directions; distance is the straight-line distance to the nearest path cell. Relief is the elevation range in a 510 m window. Profile and plan curvature follow Zevenbergen and Thorne [25]; both are positive on convex ground (spurs, fan surfaces) and negative on concave ground (hollows). The wetness index is $\ln(a/\tan\beta)$ with a the D8 specific catchment area, with small offsets for flat cells. Valley width is the width of ground with HAND ≤ 20 m, averaged over 500 m of path; channel slope is the gradient of the path over ± 500 m; bend exposure is the curvature of the smoothed path, positive on the outer bank.

B. Targets and learners

There are three binary targets: (a) affected vs. not affected (primary); (b) destroyed vs. not destroyed; (c) destroyed vs. damaged or possibly damaged, among affected buildings only. Nine classical learners were compared: decision tree, k-nearest neighbours, elastic-net logistic regression, shrinkage LDA, Gaussian naive Bayes, linear SVM, RBF SVM (Nyström approximation), random forest and histogram gradient boosting; such learners remain competitive with deep learning on small tabular data [26]. Linear regression fits the attenuation curve, and K-means and Ward clustering look for damage regimes. Features are standardised for all but the tree-based learners, and every learner that supports it uses class weights (grids in Table XVIII). The fixed destroyed | affected models use the attenuation set, the target they help most (Section V-F).

C. Class imbalance

Prevalence differs strongly by reach (Table II). Five strategies were compared: no correction, class weights, SMOTE [15] and ISMOTE [27], each at a 0.5 threshold, and no correction with a threshold tuned by Youden's J on out-of-fold training predictions. ISMOTE repeatedly oversamples the minority points the current model misclassifies ($K = 5$, up to 20 iterations). Oversampling is fitted on the training reaches only.

D. Validation

a) Why not random splits: Neighbouring buildings share terrain and flood exposure, so a random split puts near-copies of each test building in the training set and overstates skill [14, 28]. Held-out buildings lie a median 6–23 km from the nearest training building, so a held-out reach is a fair stand-in for a new valley section.

b) Main test: Leave-one-reach-out: train on two reaches, test on the third. Hyperparameters are tuned in an inner 3-fold cross-validation within the training reaches, grouped by 1-km spatial blocks (grid search on ROC-AUC), so no label of the held-out reach is used to score any choice; the decision threshold is set by Youden's J on out-of-fold training predictions. The final evaluation and the secondary analyses use fixed hyperparameters (logistic regression $C = 1$ with class weights, linear SVM $C = 0.1$, random forest with 300 trees, depth 16, minimum leaf 5 and $\sqrt{p}$ features, and gradient boosting with learning rate 0.1 and 15 leaves), so that the models evaluated are the ones interpreted later. The tree settings are the most frequent choice of the inner grid search for the affected target over the three splits. Because each reach is a training reach in two of the three splits, the setting used to score a reach is indirectly influenced by that reach's labels through the other two searches; the per-split tuned results of

Section V-D are the clean estimate, and for gradient boosting, whose fixed settings score higher than its tuned ones, the fixed scores are an upper bound. Held-out AUCs have 95% intervals from a block bootstrap (1-km blocks of the held-out reach, 1,000 resamples), with models compared on the same resamples.

c) Inner folds: Two designs were compared: 1-km spatial blocks, and k-fold nearest-neighbour distance matching (kNNDM) [29, 30], whose folds match the distances from the held-out reach to the training reaches. The *optimism gap* is the inner-CV AUC minus the held-out AUC; part of it is the optimism of selecting the best grid point.

d) Near-channel test: Most buildings stand far above any flood, and a model that merely separates valley floor from hillside scores well on them. Every model is therefore also scored on the near-channel subset: the buildings within 300 m of the flood path and below a reach-specific HAND limit, 30 m in the middle and lower reaches, about the middle-reach damaging height and a generous limit in the lower reach, and 60 m in the upper reach, about its damaging height for the destroyed target (Section V-G). The upper reach needs its own limit because every one of its buildings below 30 m was destroyed, so a 30 m subset would hold no negative for any target there. The models are trained on all buildings, as they would be for a new valley, and only scored on the subset; the destroyed | affected target is scored on the affected buildings of the subset. Counts are given in Table IX, and the limits are varied in Section V-C.

e) Area of applicability: Following Meyer and Pebesma [28], a dissimilarity index (DI) measures each held-out building's distance to the nearest training building in the standardised feature space, weighted by permutation importance. Buildings whose DI exceeds the upper whisker of the cross-validated training DI lie outside the area of applicability (AOA).

f) Metrics: Prevalence ranges from 9% to 77% between reaches, so accuracy would mostly reflect prevalence. ROC-AUC, which measures ranking and does not depend on prevalence, is the main metric; the area under the precision–recall curve (PR-AUC), which does, shows how useful a ranking is when damage is rare, and is interpreted alongside the denominator range of Table V, as are the affected shares of Section V-I. Balanced accuracy, precision and recall at the tuned threshold describe the map a user would receive.

g) Circularity: The debris flood followed the valley floor, and the photo-interpreters graded the buildings inside the trace they saw, so a terrain model predicts flood extent as well as damage given exposure. The near-channel test reduces the dominance of hillside negatives, and the destroyed | affected target asks only about severity among buildings the flood reached.

h) Calibration: We also report the Brier score, the calibration intercept (calibration-in-the-large; 0 is ideal), the calibration slope (1 is ideal) and the expected calibration error (ECE), since imbalance corrections distort probabilities [16, 31]. The probability-level analyses (recalibration, the one-settlement test and the affected shares of Section V-I) therefore use logistic regression *without* class weights, whose probabilities are meant to be read as such; class weights shift the intercept by design, so the class-weighted model of Table VIII is recalibrated beside it (Table XIII). To test cheap recalibration, the intercept alone is re-fitted on k randomly chosen buildings of the held-out reach (k = 10–200, 200 repeats; the affected share of Section V-I uses 100 draws of $k = 50$) and the rest are scored. Because the first labels after an event tend to come from one accessible settlement, the re-fit was repeated with the k = 20 or 50 buildings nearest a random seed building (one settlement, about 100–300 m across).

E. Attenuation curve

Buildings were binned in 3-km steps of along-river distance. In each bin with at least 15 buildings of each class, a one-feature logistic regression on HAND gives HAND_{50}, the height at which the damage probability is 50%. HAND_{50} was regressed on distance d from the border, linearly and in $\log(d + c)$, with the offset c chosen by leave-one-bin-out cross-validation from $\{1, 2, 5, 10, 20, 40\}$ km; c is a fitted shape parameter, separate from the fixed 10 km offset of the model feature. The slope is reported as the change per doubling of $d + c$. Its 95% interval comes from 300 bootstrap resamples of 1-km blocks (form and c fixed), because neighbouring buildings are not independent; resampling buildings and resampling bins are reported for comparison.

F. Interpretation

HAND and distance are correlated (Spearman $\rho = 0.67$), so permutation importance and partial dependence are read with care [32]. Four views are combined: standardised logistic-regression coefficients with 95% intervals from 300 block-bootstrap resamples; permutation importance on the held-out reach (mean drop in ROC-AUC over 10 shuffles); partial dependence; and a depth-3 decision tree (at least 100 buildings per leaf). The held-out errors are mapped, and a Mantel–Haenszel test within strata of reach, HAND and distance relates them to forest cover from the National Land Cover Monitoring System (NLCMS) 2022 of the Forest Research and Training Centre (FRTC) and ICIMOD [23].

G. Implementation

Python 3.12 with scikit-learn [33], imbalanced-learn [34] and pysheds [35]; exact versions are pinned in the repository, and seeds are fixed. ISMOTE, kNNDM and the AOA were re-implemented in Python following the published methods, as no code was available to us. One script rebuilds every result and figure from the raw downloads (Section VII, Data and code availability).

IV. Experimental design

The study is organised as nine experiments on one building table, listed in Table VII. E1 (RQ1), E5 (RQ2), E3 (RQ3) and E6 (RQ4) answer the research questions directly; E2, E4 and E7–E9 test the parts of the pipeline on which those answers rest. An exploratory analysis (Section V-A, not numbered in the table) precedes them: single-feature AUCs, correlations, PCA and feature selection on all buildings, from which the main feature set was fixed.

TABLE VII
THE NINE EXPERIMENTS. LR: LOGISTIC REGRESSION; RF: RANDOM FOREST; GB: GRADIENT BOOSTING; MAIN / ATT.: FEATURE SETS OF TABLE VI.

Exp.	Question	Models and features	Protocol and repeats	Compared against	Decided by
E1	Transfer to an unseen reach (RQ1)	Four fixed models (LR, linear SVM, RF, GB), with all nine learners also tuned; main set (att. set for the fixed destroyed | affected models); three targets	Train on two reaches, score the third; 1,000 block-bootstrap resamples of the held-out reach; paired differences on the same resamples	Fixed models vs. LR (paired); tuned learners by mean AUC	ROC-AUC; PR-AUC, balanced accuracy, recall and precision at the tuned threshold
E2	What terrain adds beyond the valley floor	Fixed models of E1, trained on all buildings of the training reaches; three targets	Scored on the 3,996 near-channel buildings (upper 604 below 60 m HAND; middle 2,308 and lower 1,084 below 30 m; all within 300 m); bootstrap as in E1; limits varied over HAND $\leq$ 10–30 m and distance $\leq$ 100–300 m in the middle and lower reaches	Label-free HAND ranking; label-fitted HAND cut chosen on the training reaches	ROC-AUC, PR-AUC; balanced accuracy for the cut
E3	Which learners transfer, and why (RQ3)	Nine tuned learners (optimism gap); fixed LR, RF and GB (kNNDM); four fixed models of E1 (AOA); main set, and att. set for a second AOA	Inner 3-fold CV with 1-km blocks for all nine learners and with kNNDM for three; optimism gap (inner minus held-out AUC); one LR-weighted dissimilarity index and AOA per held-out reach and feature set	Inner ranking of learners vs. held-out ranking; fixed models vs. LR (E1 pairs); AUC inside vs. outside the AOA	Mean held-out AUC; optimism gap; AUC inside vs. outside the AOA
E4	Which terrain groups add information	Fixed LR ($C = 1$) and fixed RF (400 trees, min. leaf 10); core, +local, +geometry, +attenuation (replaces geometry), all (geometry + attenuation)	Mean held-out AUC over the three reaches; near-channel models trained within the subset	Core features (HAND and distance)	Mean ROC-AUC per group
E5	Attenuation of the damaging height (RQ2)	One-feature LR on HAND per 3-km bin; least squares on HAND_{50}	Bins with $\geq$ 15 buildings per class; $c \in \{1, 2, 5, 10, 20, 40\}$ km by leave-one-bin-out CV; 300 resamples under three schemes (1-km blocks, buildings, bins)	Linear vs. $\log(d + c)$ form	Leave-one-bin-out RMSE; slope per doubling with 95% interval
E6	What transfers: the ranking or the probabilities (RQ4)	LR (no correction, class weights, SMOTE), RF, GB; main set	Calibration per held-out reach (all five); intercept re-fitted on k = 10–200 random labels, 200 repeats (unweighted and class-weighted LR); 20 or 50 labels from one settlement, 200 repeats; affected share with 50 labels, 100 draws (unweighted LR)	Uncorrected model; observed affected share	Brier, intercept, slope, ECE; share error in percentage points
E7	Class imbalance	LR and RF; no correction, tuned threshold, class weights, SMOTE, ISMOTE ($K = 5$, up to 20 iterations); affected, main set	Oversampling fitted on the training reaches only; mean over held-out reaches	No correction at a 0.5 threshold	ROC-AUC, balanced accuracy, F1, Brier
E8	Robustness to inputs and labels	Four-feature LR (inputs); fixed LR (labels)	Channel definition (flood path; rivers $\geq$ 200, 500, 1,000 km^2), GLO-30 vs. FABDEM, OpenStreetMap-only negatives (also with the fixed model); label rules R1 and R2, possibly damaged removed, lower-reach destroyed recoded, and a bound recoding 25–66% of the lower reach's destroyed buildings	Main configuration	Mean held-out ROC-AUC
E9	How the models decide; damage regimes	Fixed LR and RF on all buildings (main set; att. set for destroyed | affected coefficients and RF importance); depth-3 tree; K-means on the 3,360 affected buildings, Ward on a 2,500-building sample	Coefficients with 300 block-bootstrap resamples; permutation importance on held-out reaches (10 shuffles); partial dependence; error map; Mantel–Haenszel canopy test; silhouette for k = 2–6	LR vs. RF; GLO-30 vs. FABDEM for canopy	Direction and size of effects; odds ratios; silhouette, adjusted Rand index

a) Common setup: E1–E4 and E6–E7 share the held-out protocol, the reference model, the seeds and the block bootstrap of Section III, so that differences between them come from what each one varies; E5 and E9 fit all buildings, and E8 uses a four-feature model for its input variants. Hyperparameters are tuned only in E1 and E3; every other experiment uses the fixed models. The main feature set is the default; the attenuation set is used for the fixed destroyed | affected models and for a second area-of-applicability check.

b) Baselines and decision rules: Each experiment has a stated comparator (Table VII). The reference model is class-weighted logistic regression; in E1–E3 the other fixed models are reported as paired differences from it on the same bootstrap resamples. E2 adds two simple baselines: a ranking by HAND alone, and a single HAND cut chosen by Youden's J on the near-channel buildings of the training reaches. A difference is called clear only when its 95% block-bootstrap interval excludes zero (E1, E2, the fixed-model pairs of E3 and the slope of E5); comparisons without intervals (the tuned-learner means of E3, E4, E7, E8 and E9) are descriptive.

V. RESULTS

A. Exploratory analysis

Affected and unaffected buildings differ most in height and distance. Across reaches the median HAND of destroyed buildings is 14–35 m, of damaged buildings 18–52 m and of unaffected buildings 53–90 m, rising upstream for every grade (Fig. 3): the share destroyed falls from near one to near zero across a HAND band at 10–20 m in the lower reach, 15–30 m in the middle and 40–60 m in the upper gorge. On their own, HAND and distance rank affected buildings with AUC 0.933 and 0.914 (all data; AUC taken as $\max(\mathrm{AUC}, 1 - \mathrm{AUC})$); channel slope, profile curvature, slope and relief reach 0.63–0.66, bend exposure 0.59, and the other features are close to 0.5.

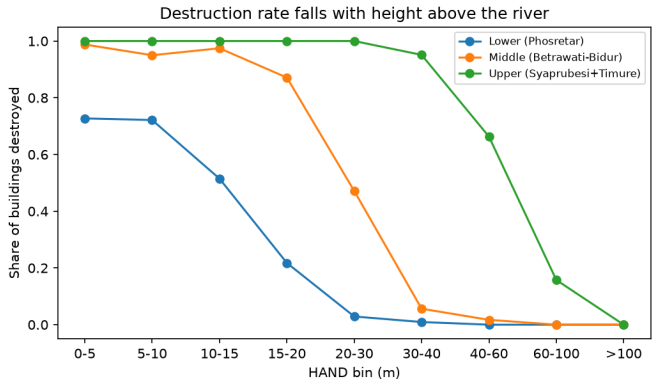


Fig. 3. Share of buildings destroyed by HAND band in each reach. The band in which destruction drops off sits highest in the upper reach and falls downstream; Section V-G fits this as the attenuation curve.

Five pairs are strongly correlated (Spearman $|\rho| \geq 0.6$), among them HAND and distance (0.67) and along-river distance and channel slope (-0.76). PCA needs four components for 67% of the variance, and three feature-selection methods agree only that HAND ranks first or second, so the main set keeps all ten physically motivated features and leaves the weighting to regularisation.

B. Transfer to an unseen reach (E1; RQ1)

Terrain alone separates affected from unaffected buildings in every held-out reach (Table VIII). Class-weighted logistic regression reaches AUC 0.970, 0.983 and 0.958 for affected and 0.952, 0.983 and 0.958 for destroyed. The upper reach has the widest intervals: it has only 12 blocks and 192 undamaged buildings. Separating destroyed from damaged among affected buildings is harder (0.930, 0.818 and 0.750).

PR-AUC shows the effect of prevalence: for affected it is 0.991, 0.929 and 0.639 at prevalences of 0.77, 0.20 and 0.09, and for destroyed it falls to 0.406 in the lower reach, where only 4.4% of the buildings in our table were destroyed (6.1% with the Copernicus inventory as denominator). At the tuned threshold, balanced accuracy for affected is 0.89, 0.94 and 0.88. The lower-reach map finds almost every affected building (recall 0.98) but only 30% of its flags are correct, while the upper-reach map misses 14% of affected buildings with 97% of its flags correct; Section V-I returns to these opposite errors.

C. Near the channel: what terrain adds beyond the valley floor (E2)

Among the near-channel buildings (Table IX), logistic regression ranks affected buildings in an unseen reach with AUC 0.851 (95% interval 0.77–0.92) in the middle reach and 0.801 (0.69–0.89) in the lower reach, and destroyed buildings with 0.859 and 0.781. In the upper reach, at its own damaging height (HAND ≤ 60 m), it ranks affected buildings with 0.881 (0.78–0.95, resting on 18 unaffected buildings), destroyed with 0.883 (0.78–0.97) and destroyed | affected with 0.931 (0.85–0.97); downstream the severity ranking is weaker, 0.800 (0.76–0.83) in the middle reach and 0.718 (0.63–0.82) in the lower.

TABLE VIII
HELD-OUT ROC-AUC OF THE FIXED MODELS (MAIN FEATURE SET; ATTENUATION SET FOR DESTROYED | AFFECTED), WITH 95% BLOCK-BOOTSTRAP INTERVALS FOR LOGISTIC REGRESSION. BOLD: BEST PER TARGET AND REACH. FIXED SETTINGS WERE SELECTED FROM INNER CROSS-VALIDATION OVER ALL THREE SPLITS (SECTION III), SO EVERY FIXED ROW MAY BE MILDLY OPTIMISTIC; THE PER-SPLIT TUNED MEANS OF TABLE X ARE THE CLEAN ESTIMATES, AND FOR GRADIENT BOOSTING THE GAP IS LARGE (SECTION V-D). PR-AUC ROWS ARE FOR THE SAME LOGISTIC REGRESSION, WITH THE POSITIVE-CLASS PREVALENCE; THE BALANCED ACCURACY ROW IS AT ITS TUNED THRESHOLD. *R1 incl.*: LOGISTIC REGRESSION WITH THE 1,282 AMBIGUOUS IN-TRACE ROWS KEPT AS POSSIBLY DAMAGED (SECTION V-K; NOT BOLDED). †THE WHOLE LOWER-REACH COLUMN FOR DESTROYED AND DESTROYED | AFFECTED IS EXPLORATORY, FOR EVERY MODEL, BECAUSE THE LOWER REACH'S DESTROYED LABELS ARE DOUBTFUL (SECTION V-K); NO BEST VALUE IS MARKED THERE.

Target	Model	Upper	Middle	Lower
Affected	LR	**0.970** (0.90–1.00)	**0.983** (0.97–0.99)	**0.958** (0.94–0.97)
	Linear SVM	**0.970**	**0.983**	0.956
	Random forest	0.858	0.957	0.947
	Grad. boosting	0.879	0.966	0.944
	LR, R1 incl.	*0.974*	*0.984*	*0.963*
	PR-AUC (LR) / prev.	0.991 / 0.77	0.929 / 0.20	0.639 / 0.09
	Bal. accuracy (LR)	0.89	0.94	0.88
Destroyed	LR	0.952 (0.88–0.99)	**0.983** (0.97–0.99)	0.958† (0.93–0.98)†
	Linear SVM	**0.953**	**0.983**	0.958†
	Random forest	0.904	0.960	0.951†
	Grad. boosting	0.943	0.963	0.954†
	LR, R1 incl.	*0.657*	*0.953*	*0.924*†
	PR-AUC (LR) / prev.	0.969 / 0.63	0.911 / 0.16	0.406 / 0.04†
Destr. | aff.	LR	**0.930** (0.87–0.98)	**0.818** (0.78–0.85)	0.750† (0.68–0.82)†
	Linear SVM	**0.930**	**0.818**	0.749†
	Random forest	0.885	**0.818**	0.713†
	Grad. boosting	0.801	0.796	0.706†
	LR, R1 incl.	*0.480*	*0.638*	*0.542*†
	PR-AUC (LR) / prev.	0.983 / 0.81	0.944 / 0.79	0.713 / 0.49†

These scores are clearly better than chance, although 12–20% of affected–unaffected pairs are still mis-ordered, and the intervals are up to about seven times wider than over all buildings (1.7, 6.3 and 7.2 times in the upper, middle and lower reaches, from the unrounded intervals). PR-AUC follows prevalence: 0.968 in the middle reach (85% affected) and 0.670 in the lower reach (40%).

Near the channel the models are closer together than over all buildings. Of the 27 paired differences between logistic regression and the other three models (positive when logistic regression is ahead), seven exclude zero: against gradient boosting in the middle reach ($+0.056$ affected, $+0.038$ destroyed) and in the upper reach for affected ($+0.17$) and destroyed | affected ($+0.14$); against random forest for affected in the upper reach ($+0.18$), where both tree models fall to 0.70–0.72; and two trivial differences against the linear SVM in the upper reach (-0.006 affected, $+0.001$ destroyed | affected). Random forest is nominally ahead of logistic regression in two cells (middle, affected and destroyed | affected, -0.009). Near the flood path the models rank buildings about equally well downstream; in the gorge the tree models fall behind even near the flood path for the affected target, which is the extrapolation

failure of Section V-E seen at close range.

Two simple baselines give a scale for these scores. Ranking by HAND alone gives AUC 0.77–0.91 for affected, 0.83–0.89 for destroyed and 0.72–0.88 for destroyed | affected, not detectably worse than the ten-feature model in any cell (paired differences -0.045 to $+0.054$, all intervals including zero). Differences under about 0.05 could not be detected, so the data show no difference but cannot show equivalence. A single HAND cut chosen on the training reaches (14–23 m) is much weaker: its AUC is 0.59–0.68, 0.10–0.30 below logistic regression, and the paired difference excludes zero in eight of the nine cells (not for destroyed in the lower reach).

a) Other cutoffs: The 30 m and 300 m limits are not neutral. In the middle reach the affected AUC stays at 0.85–0.88 for every combination of limits with at least 15 buildings of each class. In the lower reach it falls to 0.64–0.76 at tighter limits, so the main limits are the most favourable ones there, and for destroyed the ranking reverses among the lowest buildings (AUC 0.25–0.30 for HAND $\leq$ 10 m, 71–86 buildings, that is, 0.70–0.75 with the sign flipped). HAND alone matched or beat the model in 18 of the 23 limit–target–reach combinations with at least 15 buildings of each class. The middle-reach result therefore holds across cutoffs; the lower-reach one does not.

D. Which learners transfer (E3; RQ3)

With the main feature set, the linear models transfer best (Table X); with the attenuation set the tree ensembles are competitive (Section V-F). On the same bootstrap samples, the fixed logistic regression of Table VIII beats the fixed random forest by $+0.112$ (95% interval $+0.022$ to $+0.206$) and gradient boosting by $+0.091$ ($+0.019$ to $+0.183$) in the upper reach, and by $+0.026$ and $+0.017$ in the middle reach; in the lower reach the intervals include zero. These are the fixed-model differences, and for gradient boosting the fixed model is the optimistic reading (see below). The fixed logistic regression and linear SVM are equal on average (0.970 for both; lower reach 0.958 vs. 0.956); the only paired difference whose interval excludes zero is $+0.002$ in the lower reach.

The order of the learners is almost the same for destroyed (linear models 0.964, decision tree and k-NN last at 0.89–0.90) and for destroyed | affected, where all learners are weaker (Table X; with the attenuation set the tuned logistic regression averages 0.827 (not tabulated), against 0.833 for the fixed model of Table VIII). LDA and naive Bayes, which assume Gaussian classes, trail logistic regression by 0.02–0.04 for affected, as the right-skewed HAND and distance distributions (skewness 1.2 and 0.9) would predict. The per-split tuned models of Table X and the fixed models of Table VIII give the same mean affected AUC for logistic regression (0.969 vs. 0.970, although the selected penalty ranged from C = 0.01 to 10 over the 27 tuned fits of three targets, three feature sets and three held-out reaches) and for the random forest (0.920 under both), so for these the fixed settings add no skill. Gradient boosting is the exception: its fixed settings score higher than the tuned ones (0.929 vs. 0.887), because the inner search chose a learning rate of 0.05 or 31 leaves in two of the three splits and those choices transfer worse, so its fixed-model scores are the more favourable reading.

TABLE IX

NEAR-CHANNEL TEST: HELD-OUT ROC-AUC FOR BUILDINGS WITHIN 300 M OF THE FLOOD PATH AND BELOW THE REACH'S HAND LIMIT (60 M UPPER, 30 M MIDDLE AND LOWER); FIXED MODELS TRAINED ON ALL BUILDINGS OF THE OTHER TWO REACHES; 95% BLOCK-BOOTSTRAP INTERVALS FOR LOGISTIC REGRESSION AND HAND ONLY. *HAND only*: RANKING BY HAND, NO MODEL. *HAND cut*: ONE THRESHOLD CHOSEN BY YOUDEN'S J ON THE NEAR-CHANNEL BUILDINGS OF THE TRAINING REACHES (UPPER / MIDDLE / LOWER); A BINARY RULE'S AUC EQUALS ITS BALANCED ACCURACY. FINAL ROW OF EACH BLOCK: PR-AUC OF LOGISTIC REGRESSION AND POSITIVE-CLASS PREVALENCE. n = 604 (UPPER), 2,308 (MIDDLE), 1,084 (LOWER); FOR DESTROYED | AFFECTED, THE 586, 1,968 AND 432 AFFECTED BUILDINGS AMONG THEM. THE UPPER-REACH COLUMNS REST ON 18 UNAFFECTED, 99 NOT DESTROYED AND 81 DAMAGED BUILDINGS. †THE WHOLE LOWER-REACH DESTROYED AND DESTROYED | AFFECTED COLUMNS, MODELS AND BASELINES ALIKE, ARE EXPLORATORY (DOUBTFUL LOWER-REACH DESTROYED LABELS); NO BEST VALUE IS MARKED THERE.

Target	Model	Upper	Middle	Lower
Affected	LR	0.881 (0.78–0.95)	0.851 (0.77–0.92)	**0.801** (0.69–0.89)
	Linear SVM	0.887	0.850	0.792
	Random forest	0.703	0.860	0.756
	Grad. boosting	0.715	0.794	0.742
	HAND only	**0.906** (0.88–0.94)	**0.876** (0.84–0.91)	0.770 (0.67–0.84)
	HAND cut (21 / 16 / 23 m)	0.614	0.652	0.644
	PR-AUC (LR) / prev.	0.996 / 0.97	0.968 / 0.85	0.670 / 0.40
Destroyed	LR	0.883 (0.78–0.97)	0.859 (0.81–0.91)	0.781† (0.66–0.90)†
	Linear SVM	0.886	**0.861**	0.783†
	Random forest	0.839	0.854	0.755†
	Grad. boosting	0.875	0.821	0.786†
	HAND only	**0.889** (0.67–0.94)	0.854 (0.83–0.88)	0.827† (0.76–0.90)†
	HAND cut (21 / 15 / 22 m)	0.633	0.647	0.683†
	PR-AUC (LR) / prev.	0.974 / 0.84	0.933 / 0.71	0.424 / 0.21
Destr. | aff.	LR	**0.931** (0.85–0.97)	0.800 (0.76–0.83)	0.718† (0.63–0.82)†
	Linear SVM	0.930	0.800	0.716†
	Random forest	0.886	**0.810**	0.665†
	Grad. boosting	0.795	0.787	0.661†
	HAND only	0.877 (0.79–0.94)	0.797 (0.74–0.83)	0.723† (0.63–0.84)†
	HAND cut (21 / 14 / 22 m)	0.633	0.628	0.591†
	PR-AUC (LR) / prev.	0.988 / 0.86	0.951 / 0.83	0.732 / 0.54

Inside the training reaches, flexible models look as good as linear ones (inner AUC for affected: random forest 0.975, logistic regression 0.974), but on a new reach they lose far more: 0.055 (random forest) and 0.087 (gradient boosting) against 0.006 for logistic regression. kNNDM folds come closer to the transfer setting than 1-km blocks (median fold-to-training distance 1.2–3.0 km instead of 250–330 m) and narrow the mean absolute optimism gap from 0.034 to 0.025, but they cannot reproduce the 6–23 km jump to a new reach, and no inner design ranks the models correctly: under kNNDM random forest still beats logistic regression inside the training reaches (0.965 vs. 0.960, means over the affected and destroyed targets), while the held-out reaches show the reverse (0.929 vs. 0.967).

E. Area of applicability (E3)

With the main feature set, 7–21% of the buildings in the middle and lower reaches lie outside the AOA, but 74–83% of those in the upper gorge do: its steep, narrow terrain is unlike anything downstream (Fig. 4). For affected, outside the AOA

TABLE X
PER-SPLIT TUNED LEARNERS, MAIN FEATURE SET: MEAN HELD-OUT AUC OVER THE THREE REACHES FOR THE THREE TARGETS (AFF.: AFFECTED; DESTR.: DESTROYED; D | A: DESTROYED AMONG AFFECTED), MEAN AFFECTED AUC OVER THE MIDDLE AND LOWER REACHES NEAR THE CHANNEL (TUNED MODELS ON THE HAND $\leq$ 30 M SUBSETS OF THOSE TWO REACHES; A DIFFERENT SUBSET FROM TABLE IX), AND THE AFFECTED OPTIMISM GAP (INNER 1-KM BLOCK CV AUC MINUS HELD-OUT AUC). BOLD: BEST PER COLUMN.

Model	Aff.	Destr.	D | A	Near-ch.	Opt. gap
Logistic regression	**0.969**	**0.964**	**0.792**	**0.818**	0.006
Linear SVM	**0.969**	**0.964**	0.781	**0.818**	0.005
RBF SVM	0.954	0.956	0.730	0.788	0.018
LDA	0.945	0.907	0.769	0.759	**0.003**
Naive Bayes	0.934	0.917	0.692	0.652	0.011
Random forest	0.920	0.940	0.724	0.808	0.055
k-NN	0.918	0.899	0.695	0.622	0.037
Gradient boosting	0.887	0.943	0.651	0.781	0.087
Decision tree	0.878	0.889	0.548	0.750	0.069

in the upper reach, random forest and gradient boosting reach AUC 0.811 and 0.843, against 0.957 for logistic regression; inside it, random forest reaches 0.959. Being outside the AOA is not enough to make the trees fail, however: with the attenuation set the whole upper reach lies outside it, yet the tuned random forest reaches 0.958 there. The AOA analysis is consistent with an extrapolation failure of the main-set trees, which learn reach-specific HAND thresholds (Section V-F), but it is only part of the explanation.

F. Which terrain groups add information (E4)

Adding the feature groups one at a time (Appendix C) raises the mean affected AUC of logistic regression from 0.954 (HAND and distance) to 0.970 with local terrain and valley geometry; the attenuation terms help destroyed | affected most (0.788 $\rightarrow$ 0.833). The random forest transfers well only with the attenuation terms (0.963 for affected vs. 0.920), because without them it learns reach-specific HAND thresholds (partial dependence, Appendix F).

G. Attenuation of the flood (E5; RQ2)

The damaging height above the flood path falls with distance downstream (Fig. 5). A logarithmic form fits far better than a straight line (leave-one-bin-out RMSE 4.7 vs. 10.4 m for affected, with $c = 5$ km; 4.0 vs. 5.5 m for destroyed, with $c = 20$ km). HAND_{50} falls by 19.5 m per doubling of $d + c$ for affected buildings (95% block-bootstrap interval 16.8–21.9 m; $R^2 = 0.94$) and by 23.0 m for destroyed buildings (19.9–28.8 m; $R^2 = 0.93$), from about 87 m (affected) and 63 m (destroyed) near the border to about 16 m and 11 m at 90 km. The two intervals overlap (19.9–21.9 m), so the data do not establish that the destroyed height falls faster than the affected one. Resampling buildings instead of blocks narrows the intervals 1.5–2.2 times (Table XI), but under every scheme the slope is far from zero. The bins are uneven: two upstream bins at 2.6 and 14.7 km (220 and 452 buildings), none between about 16 and 42 km, and 16 bins (15 for destroyed) between 42 and 91 km, so the curve across the gap is interpolated and its high R^2 owes much to the two upstream bins. Fitted to the downstream bins alone, the fall has the same direction (16.9 m per doubling for affected; 27.6 m for destroyed, larger than in the full fit), but R^2 drops to 0.58 and 0.79.

TABLE XI
FALL OF HAND_{50} (M) PER DOUBLING OF $d + c$, WITH 95% INTERVALS FROM THREE BOOTSTRAP SCHEMES (300 RESAMPLES EACH; FORM AND c FIXED).

Target	Slope	1-km blocks	Buildings	Bins
Affected, c = 5	19.5	16.8–21.9	17.7–21.1	14.9–21.3
Destroyed, c = 20	23.0	19.9–28.8	21.5–25.5	20.8–30.2

c in km. All values are magnitudes of the fall in HAND_{50}.

TABLE XII
CALIBRATION (MAIN SET, FIXED MODELS OF TABLE VIII) PER TARGET: AUC, BRIER, CALIBRATION SLOPE AND ECE ARE MEANS OVER THE THREE HELD-OUT REACHES; THE INTERCEPT (CALIBRATION-IN-THE-LARGE) IS GIVEN PER HELD-OUT REACH (U / M / L); POSITIVE MEANS UNDER-PREDICTION. BOLD: LOWEST BRIER AND ECE PER TARGET.

Model	AUC	Brier	Slope	ECE	Intercept U / M / L
Affected					
LR, no correction	0.971	0.089	0.86	0.107	$+4.1$ / $+0.8$ / -1.5
LR, class weights	0.970	**0.078**	0.83	**0.090**	$+2.3$ / -0.5 / -2.6
LR, SMOTE	0.970	0.083	0.80	0.097	$+2.9$ / -0.6 / -2.6
Random forest, weights	0.920	0.131	1.51	0.169	$+1.8$ / -0.1 / -2.1
Grad. boosting, weights	0.929	0.161	0.57	0.176	$+3.9$ / $+1.0$ / -3.7
Destroyed					
LR, no correction	0.965	**0.070**	0.85	**0.072**	$+2.0$ / $+0.9$ / -1.7
LR, class weights	0.964	0.072	0.74	0.075	$+0.3$ / -0.9 / -3.2
LR, SMOTE	0.965	0.073	0.68	0.075	$+1.0$ / -1.0 / -3.3
Random forest, weights	0.937	0.116	1.61	0.140	$+1.3$ / -0.4 / -2.6
Grad. boosting, weights	0.953	0.144	0.62	0.163	$+4.8$ / -0.8 / -3.8

H. Ranking transfers, probabilities do not (E6; RQ4)

The probability analyses below use logistic regression without class weights unless stated (Section III); Table XII shows that the imbalance strategy moves the probability level but not the ranking. The calibration intercept follows flood strength (Fig. 6, Table XII). Trained on the middle and lower reaches, logistic regression *under*-predicts damage in the upper reach (affected intercept +2.3 to +4.1, depending on the imbalance strategy); trained on the upper and middle reaches, it *over*-predicts in the lower reach (-1.5 to -2.6). The middle reach, between the others, is close to calibrated. The destroyed target behaves the same way, with the unweighted model best calibrated (Brier 0.070, ECE 0.072) because class weights push its rarer positives too high downstream (lower-reach intercept -3.2). Tree models are the worst calibrated for both targets: random forest probabilities are too timid (slope 1.5–1.6) and gradient boosting's too extreme (0.6).

Re-fitting only the intercept removes most of this error with few labels (Table XIII): with 20 randomly chosen labelled buildings, the ECE of the unweighted model falls from 0.211 to 0.047 in the upper reach and from 0.074 to 0.028 in the lower reach. The class-weighted model of Table VIII starts from a different error, because the weights lift every probability: this happens to help in the upper reach (ECE 0.107 before re-fitting) and to hurt in the lower reach (0.140), but after re-fitting on 50 or more labels both models sit at the same level (0.038–0.040

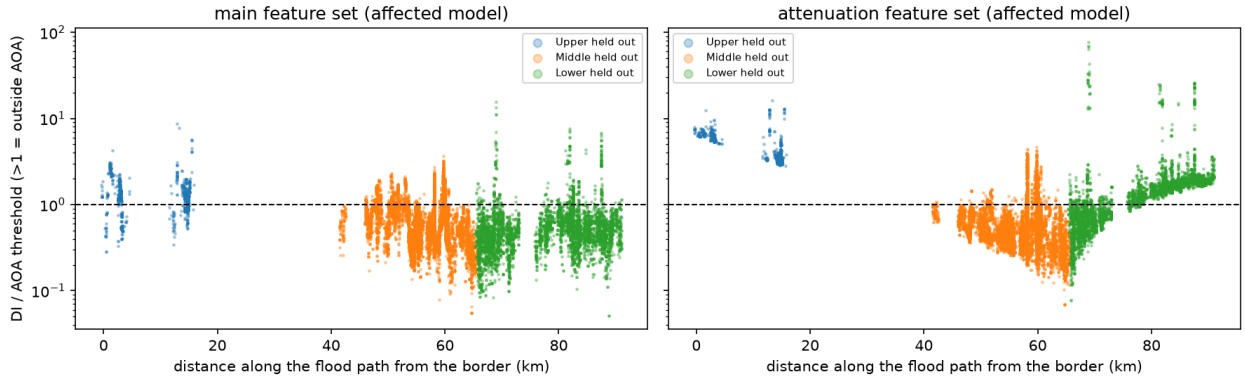


Fig. 4. Dissimilarity index of held-out buildings relative to the AOA threshold (values above 1 lie outside the area of applicability), against distance along the flood path. Left: main feature set. Right: attenuation set, where along-river distance is extrapolated for the upper and lower reaches.

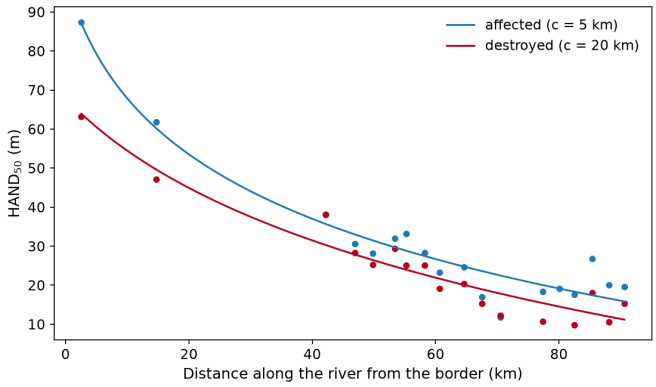


Fig. 5. Attenuation curve. HAND_{50} per 3-km bin against distance along the flood path from the border, with the fitted logarithmic curve in $d + c$ (c = 5 km for affected, 20 km for destroyed). Fall per doubling of $d + c$: 19.5 m for affected (95% block-bootstrap interval 16.8–21.9) and 23.0 m for destroyed (19.9–28.8). No buildings lie between 16 and 42 km.

TABLE XIII
INTERCEPT-ONLY RECALIBRATION WITH k LABELLED BUILDINGS OF THE HELD-OUT REACH: ECE OF THE AFFECTED TARGET ($k = 0$: BEFORE RECALIBRATION; MEDIAN OF 200 REPEATS; 10TH–90TH PERCENTILES FOR THE UNWEIGHTED MODEL IN SMALL TYPE) FOR LOGISTIC REGRESSION WITHOUT CORRECTION (*none*) AND FOR THE CLASS-WEIGHTED MODEL OF TABLE VIII (*weights*).

Reach	Model	0	10	20	50	100	200
Upper	none	0.211	0.073	0.047	0.040	0.039	0.038
			.035–.141	.034–.101	.034–.065	.034–.051	.032–.046
	weights	0.107	0.047	0.044	0.040	0.039	0.038
Middle	none	0.036	0.031	0.024	0.019	0.015	0.013
			.010–.078	.010–.059	.010–.043	.010–.031	.009–.022
	weights	0.023	0.026	0.026	0.022	0.016	0.014
Lower	none	0.074	0.040	0.028	0.025	0.022	0.022
			.020–.086	.019–.071	.019–.049	.019–.032	.019–.030
	weights	0.140	0.049	0.031	0.027	0.024	0.023

upper, 0.022–0.027 lower). The spread over draws is wide (10th–90th percentiles in Table XIII): one unlucky draw of 20 can leave half of the original error in the upper reach and nearly all of the smaller lower-reach error.

TABLE XIV
ERROR OF THE PREDICTED AFFECTED SHARE (PERCENTAGE POINTS, MEDIAN OF 200 REPEATS; 10TH–90TH PERCENTILES IN SMALL TYPE) AFTER RE-FITTING THE INTERCEPT ON k LABELS DRAWN AT RANDOM OR FROM ONE SETTLEMENT; LOGISTIC REGRESSION WITHOUT CORRECTION.

		$k = 20$		$k = 50$	
Reach	**None**	**Random**	**One settl.**	**Random**	**One settl.**
Upper	21.1	4.3	9.3	2.4	5.4
		0.6–10.6	1.3–21.1	0.5–5.3	0.8–12.7
Middle	3.6	2.2	3.6	1.4	3.6
		0.6–5.6	3.6–8.9	0.4–3.8	1.9–7.7
Lower	7.4	2.8	6.5	2.0	5.3
		0.5–6.8	1.5–7.4	0.4–4.0	1.8–7.4

I. *Predicted versus observed affected share (E6)*

For a responder, the first question after an event is often how many buildings were hit. Without recalibration, the unweighted logistic regression predicts 56% affected instead of 77% in the upper reach and 17% instead of 9% in the lower reach (Table XV, Fig. 7): it pulls every reach towards the average flood of its training reaches. After recalibration on one draw of 50 randomly spread labels, an idealised spread (the one-settlement test below is the realistic case), the predicted share is off by a median of 1.4–2.4 percentage points (Table XIV); averaged over 100 draws it is within one percentage point of the share in our table in all three reaches. The thresholded map is worst: in the lower reach it flags 1,835 buildings, 3.2 times the 566 recorded as affected (30% of the table, against a recorded 9–13.5% depending on the denominator; Table V), although its ranking there is good (AUC 0.958).

With labels from one settlement instead, the gain is smaller and uneven (Table XIV): the upper reach still benefits (share error 21.1 $\rightarrow$ 5.4 percentage points with 50 labels, against 2.4 for random labels), the lower reach benefits less (7.4 $\rightarrow$ 5.3, against 2.0), and in the middle reach the median error is unchanged (3.6 under both), although the percentiles in Table XIV show that a single settlement can help or hurt there. The cheap fix therefore needs labels from several settlements, especially downstream.

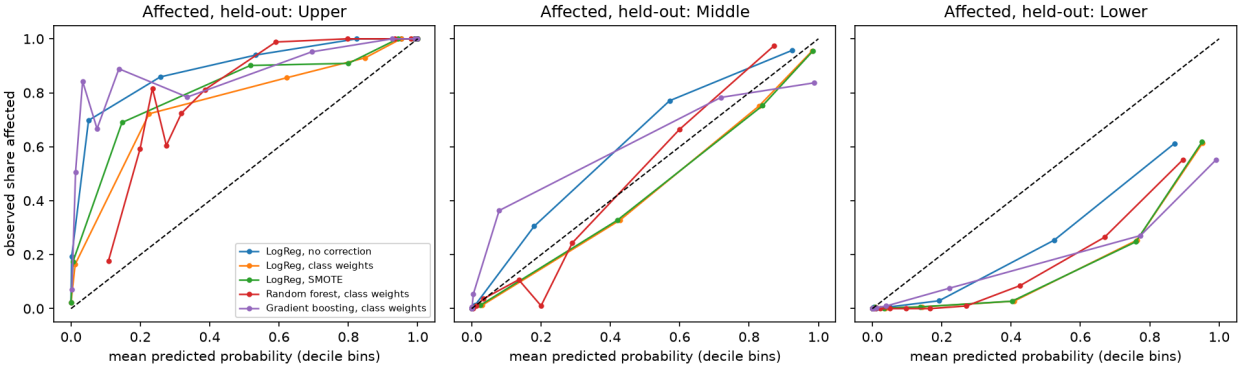


Fig. 6. Reliability diagrams for the affected target in each held-out reach (decile bins). Points above the diagonal are under-predicted; below it, over-predicted.

TABLE XV
SHARE OF BUILDINGS AFFECTED PER HELD-OUT REACH: OBSERVED, PREDICTED (LOGISTIC REGRESSION, NO CORRECTION), RECALIBRATED ON 50 RANDOM LABELS (MEAN PREDICTION OVER 100 DRAWS; A SINGLE DRAW IS OFF BY A MEDIAN OF 1.4–2.4 POINTS, TABLE XIV), AND FLAGGED AT THE TUNED THRESHOLD (CLASS-WEIGHTED MODEL). DENOMINATOR: OUR TABLE (LOWER-REACH RANGE 9.1–13.5% IN TABLE V).

Reach	n	Observed	Predicted	Recal.	Flagged
Upper	845	0.773	0.562	0.778	0.683
Middle	10,456	0.205	0.169	0.201	0.241
Lower	6,202	0.091	0.166	0.092	0.296

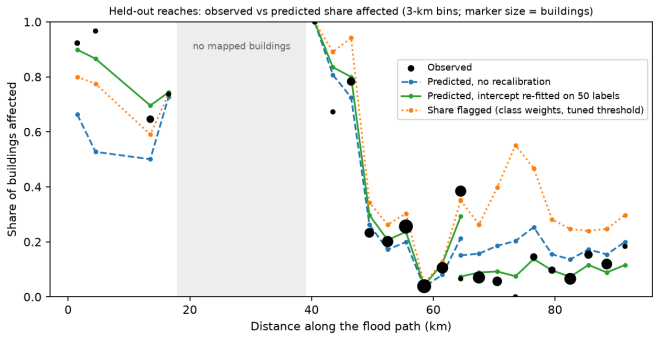


Fig. 7. Observed and predicted share of affected buildings per 3-km bin of the held-out reaches (bins with at least 30 buildings; marker size shows the number of buildings).

J. Class imbalance (E7)

Class weights, SMOTE and threshold tuning leave the ranking essentially unchanged (mean AUC 0.970–0.971), while ISMOTE is worse for logistic regression on every measure (AUC 0.939). Appendix B (Table XIX) gives the thresholded metrics.

K. Sensitivity (E8)

With a reduced, four-feature logistic regression, the channel definition, the elevation model (GLO-30 or FABDEM) and the source of the negatives change the mean affected AUC by at most 0.002, and destroyed by at most 0.011 (Appendix D, Table XXI). With the fixed ten-feature model, OpenStreetMap-only negatives change the mean AUC by less than 0.01 (Appendix D).

The label decisions were then varied for the fixed model (main set for affected and destroyed, attenuation set for destroyed | affected; Table XVI). The affected target hardly moves under any variant (0.970–0.981); switching the footprint-size rule R2 off changes nothing, and raising its minimum to 20 m^2 moves affected by 0.001 and destroyed by 0.003. Removing the noisiest grade, *possibly damaged*, lowers destroyed | affected (0.833 $\rightarrow$ 0.777), because the remaining damaged class is smaller. The ambiguous-row rule R1 has the largest effect: relabelling the 1,282 R1 rows as possibly damaged drops destroyed in the upper reach to 0.657, and destroyed | affected to a three-reach mean of 0.553, with 0.480, below chance, in the upper reach. Those rows lie inside the flood trace, as low as the destroyed buildings (median HAND 18 m vs. 18.7 m), so counting them as merely possibly damaged places low-lying buildings in the non-destroyed class. Excluding them, as in the main analysis, is therefore optimistic for the destroyed targets. Table VIII reports both readings; the affected results hold under either.

The audit doubt about the lower reach's destroyed labels was tested by recoding its 275 "destroyed" buildings as damaged; the lower reach then has no destroyed buildings left to score. In the other two reaches destroyed changes by at most 0.025 and destroyed | affected by at most 0.035, so the doubtful labels do not drive the upstream results. Within the lower reach, however, they do. Recoding the audit's share of standing buildings (8 of 18; Wilson 95% interval 25–66%) at random hardly moves the all-building destroyed AUC (0.944–0.953) but lowers PR-AUC from 0.41 to 0.14–0.31; if the recoded buildings are those the model scores highest, or lowest, the near-channel destroyed AUC ranges from 0.44 to 0.95 and destroyed | affected from 0.41 to 0.89. The lower-reach destroyed and destroyed | affected results are therefore exploratory; the severity claims rest on the upper and middle reaches.

L. Damage regimes (E9)

K-means and Ward clustering of the 3,360 affected buildings find only one weak contrast, between the steep gorge and the open valley (silhouette 0.27 at $k = 2$), with a similar share

TABLE XVI
LABEL SENSITIVITY OF THE FIXED LOGISTIC REGRESSION: MEAN HELD-OUT ROC-AUC OVER THE REACHES SCORED. AFF.: AFFECTED; DESTR.: DESTROYED; D | A: DESTROYED AMONG AFFECTED (ATTENUATION SET).

Variant	Aff.	Destr.	D | A
Main (R1–R3)	0.970	0.964	0.833
R1 rows → poss. damaged	0.974	0.845	0.553
No minimum footprint (R2 off)	0.970	0.964	0.833
Minimum footprint 20 m^2	0.971	0.961	0.833
Possibly damaged removed	0.981	0.974	0.777
Lower-reach destr. → damaged[a]	0.970	0.952	0.868

[a]Affected: all three reaches; destroyed and D | A: upper and middle only, as the lower reach has no destroyed buildings left. Before recoding, the means over those two reaches are 0.967 (Destr., main set) and 0.874 (D | A, attenuation set).

destroyed in both (74% and 78%); there are no further distinct damage regimes (Appendix E, Fig. 8).

M. Interpretation (E9)

a) Coefficients: HAND dominates every target (standardised coefficient for affected -6.1, 95% block-bootstrap interval -8.6 to -4.7; all coefficients and intervals in Table XXII), followed by distance (-1.9, -2.9 to -0.9) and channel slope ($+1.2$, $+0.7$ to $+1.9$; $+1.5$ for destroyed). At equal height and distance, buildings beside steeper channels, on steep terrace edges (slope $+0.66$, $+0.42$ to $+1.04$; relief $+0.50$, $+0.17$ to $+0.95$) and on convex fan and spur surfaces (plan curvature $+0.31$, $+0.15$ to $+0.48$) were hit more often; outer-bank buildings were more often destroyed (bend $+0.25$, $+0.01$ to $+0.56$) but not more often affected ($+0.06$, -0.30 to $+0.41$). The intervals of profile curvature, valley width and the wetness index include zero for every target. Among affected buildings, only HAND (-3.8, -5.8 to -1.2) and the along-river term $\log(d + 10)$ (-3.1, -4.2 to -1.8) separate destroyed from damaged.

b) Permutation importance on unseen reaches: On the held-out reaches, shuffling HAND lowers the AUC of logistic regression by 0.37 as a mean over the three held-out reaches, distance by 0.02 and no other feature by more than 0.01 as a mean (Appendix F). HAND and distance share much of their information, so the secondary features can show small permutation values even though their coefficients above are stable [32].

c) Rules: The depth-3 tree reduces to two readable rules for *affected*: a building below 31 m of HAND close to the path, or one above 31 m beside a steep channel; the second rule is the gorge (thresholds in Appendix F). Partial dependence agrees with the coefficients on the direction of every effect (Appendix F).

d) Errors and land cover: The held-out errors follow the calibration result: the threshold learned upstream over-predicts in the lower reach and under-predicts in the upper (Appendix G). A Mantel–Haenszel test within strata of reach, HAND and distance does not support canopy in the surface model as the driver of the misses: near-forest buildings are less often missed and more often false positives, because they stand on steeper ground (odds ratios in Appendix G).

VI. DISCUSSION

a) Terrain predicts where the flood reached: Terrain alone ranks affected buildings with AUC above 0.95 in every unseen reach, with balanced accuracy 0.88–0.94 at the tuned threshold. Because the flood trace is largely controlled by terrain, these all-building scores are an upper bound that partly measures flood extent; the near-channel scores and the destroyed | affected scores (Sections V-C and V-K) are the numbers we would quote for a new valley. The near-channel pairs that remain mis-ordered are the part of damage that a 30 m elevation model cannot see. There the data are consistent with HAND carrying most of the ranking signal; what the model adds is mainly the decision threshold and the probability level, which a fixed HAND cut gets badly wrong and which depend on the event and the reach.

b) Why linear models transfer: Paulik et al. [8] and Currie and Reinke [14] found that damage models lose accuracy across locations and event types; our transfer test reproduces this for flexible models and offers a partial explanation. With the main feature set the tree models cannot extrapolate beyond the feature ranges of the training reaches, so their predictions flatten in the gorge, while logistic regression extrapolates monotone HAND and distance effects, which held there in this event. With the attenuation set, however, the trees transfer well although the upper reach lies outside their area of applicability. Inner cross-validation, even with kNNDM folds, cannot reveal this; with two training reaches, only a held-out reach can.

c) The flood weakens with distance: The damaging height falls with distance downstream as a logarithmic curve (Section V-G). Channel slope correlates with along-river distance ($\rho = -0.76$), so much of the weakening is captured by where the valley flattens and widens, which is why the geometry features transfer without extrapolation.

Wang et al. [11] found that brick–concrete buildings reach near-complete damage at a flow depth of about 2.5 m or an impact pressure of 25 kPa, and that momentum flux separates damage grades better than depth. In our data the destroyed curve lies below the affected curve by about 5 m at 90 km but by about 24 m near the border. If depth alone set the grade, this gap would stay at a few metres; its growth upstream suggests that in the confined gorge, destruction depended on velocity and impact more than on depth alone. HAND is not a flow depth, so the comparison is qualitative, but it points the same way as the finding of Wang et al. [11].

d) What generalises to another flood: The calibration results answer RQ4. The ranking of buildings transfers; the probability level does not, and its error follows the attenuation curve: a model trained downstream under-predicts the stronger flood upstream, and vice versa. Barnhart et al. [10] found the same dependence on event size for physically based debris-flow forecasts. A terrain-only model can therefore be generalised in two ways. Before an event, it supplies exposure rankings or scenario maps for several HAND_{50} levels. After an event, 20–50 buildings labelled from the first imagery fix the probability level, provided they are spread over the area; labels from a single settlement remove only part of the error. Currie and Reinke [14] likewise found large transfer losses for radar-based damage models, so the loss is not specific to terrain; our

contribution is to show that its direction follows flood strength and that a few well-spread labels correct it.

e) What the features say about the flood: The secondary coefficients (Section V-M) are small but physically consistent: steeper channels, convex fan and spur surfaces, steep terrace edges and outer banks are where a debris flood is fastest, spreads and erodes, while the features that describe slow ponding (valley width, wetness index) add nothing. Near the channel they did not detectably improve the ranking over HAND alone, within the roughly 0.05 resolution of that test, so more derivatives of the same grid may not help there; a finer elevation model that sharpens HAND itself is the more promising route.

Limitations

f) Labels: The grades come from photo-interpretation under time pressure, without field survey, and Copernicus publishes no accuracy for them. Undamaged buildings are inferred from the absence of a grade, and some footprints are not buildings, so prevalence carries a denominator range (Table V). The affected results change little under the label variants, but the destroyed results depend on the 1,282 ambiguous in-trace rows, and the lower-reach destroyed results are exploratory.

g) Circularity: Part of the skill comes from predicting flood extent; the near-channel and destroyed | affected scores are the fairer measures, and the all-building scores are an upper bound.

h) Features: Only terrain was used: building material, storeys and age, which drive earthquake damage models [12, 18], are absent, as are deposit thickness and flow velocity. The 30 m surface model includes canopy, though the error analysis does not support it as the driver of the misses (Appendix G).

i) Design: All findings come from one event; the upper reach has only 845 buildings in 12 blocks, so its intervals are wide. The decision threshold learned on two reaches is applied to a third with different prevalence, so thresholded metrics are less stable than ranking metrics, and the recalibration result assumes labels spread over the reach. The attenuation fit is unweighted across bins of 55–3,600 buildings, has no data between 16 and 42 km and only two bins upstream of that gap, and its block bootstrap treats 1-km blocks as independent. ISMOTE, kNNDM and the AOA are our own re-implementations.

VII. Conclusions

This study asked whether terrain alone can rank buildings by recorded damage from the 2026 Langtang Lirung–Trishuli debris flood, and whether such a model transfers along the valley. Class-weighted logistic regression on ten terrain features, matched by a linear SVM, ranked affected buildings with AUC 0.958–0.983 in unseen reaches and 0.85–0.88 among near-channel buildings of the upper and middle reaches (0.80 in the lower), where HAND alone was not detectably worse, and separated destroyed from damaged buildings near the channel with 0.80–0.93 in the upper and middle reaches; the lower-reach value of 0.72 is exploratory (RQ1). The fitted attenuation curve implies a damaging height that falls by 19.5 m per doubling of along-river distance for affected buildings and 23 m for destroyed buildings (RQ2); the two rates are not shown to differ, and the curve describes this event only. With the main feature set the linear models transferred best, and the tree ensembles lost most skill in the upstream gorge (RQ3). The ranking transferred but the probabilities followed flood strength; 20–50 labelled buildings spread over a reach typically recalibrated them, for the unweighted and the class-weighted model alike (RQ4).

Two further results qualify these answers. Terrain says more about whether a building was reached than about how badly it was hit: separating destroyed from damaged buildings was harder, and the destroyed results depend on how the 1,282 ambiguous in-trace rows are read (Section V-K). And a thresholded map trained upstream flagged far more lower-reach buildings than were recorded affected, although its ranking there was good.

a) Practical implications: A terrain-only model may help screen exposure in other Himalayan valleys before an event, once tested beyond this one. A hypothesis for planning is that setback heights above the river should depend on distance from likely sources. After an event, a ranking can be produced at once and recalibrated from a few labelled buildings.

b) Future research: Three tests follow: on the July 2025 Bhote Koshi outburst flood, which came down the same channel a year earlier and destroyed the Rasuwagadhi bridge, once building-level labels exist; on other cascades; and with an event-scale input in place of labels, such as a gauge stage converted through HAND rating curves [17]. A finer elevation model would show whether the lower near-channel results come from the 30 m grid, and field-checked grades would bound the label error that limits the destroyed target.

c) Data and code availability: All inputs are openly licensed (Table III); FABDEM and the Vantor imagery are for non-commercial use. One script (`run_all.sh`) rebuilds every number and figure from the raw downloads with a pinned environment, and the per-cell values behind every table, including the paired bootstrap differences, are in its results folder. The code, the cleaned building table, the audit judgements and this manuscript are at https://github.com/rustamniraula90/langtang-trishuli-damage (MIT licence for code) and are archived on Zenodo under concept DOI https://doi.org/10.5281/zenodo.23802508, which resolves to the latest version; release v1.1.2 (https://doi.org/10.5281/zenodo.23955593) is the version this manuscript reports, and v1.0.1 (https://doi.org/10.5281/zenodo.23802509) the first-submission version.

Appendix A
Data and tuning details

Table XVII gives the cleaning log of Section II, and Table XVIII the tuning grids of the nine learners.

Appendix B
Class imbalance (E7)

Table XIX gives the thresholded metrics. Without correction, a 0.5 threshold misses many affected buildings (recall 0.79);

TABLE XVII
CLEANING LOG. RULES R1–R3 REMOVE UNGRADED FOOTPRINTS ONLY.

Step	Rows	Removed
EMS points and pre-event footprints in analysed areas	18,908	–
R1: ungraded footprints inside the flood trace	17,626	1,282
R2: ungraded footprints $< 10\,\mathrm{m}^2$	17,520	106
R3: ungraded twins < 8 m from an affected footprint	17,503	17

TABLE XVIII
LEARNERS AND TUNING GRIDS (INNER 3-FOLD CV ON ROC-AUC).

Learner	Grid
Logistic regression	$C \in \{0.001, 0.01, 0.1, 1, 10\}$; L1 ratio $\in \{0, 0.5, 1\}$
LDA	shrinkage none or Ledoit–Wolf
Decision tree	depth $\in \{2, 3, 4, 6, 8\}$; min. leaf $\in \{10, 50, 200\}$
k-NN	$k \in \{5, 15, 25, 51, 101\}$; uniform or distance weights
Naive Bayes	variance smoothing $\in \{10^{-9}, 10^{-6}, 10^{-3}\}$
Linear SVM	$C \in \{0.01, 0.1, 1\}$
RBF SVM	500 Nyström components; $\gamma \in \{0.05, 0.2, 1\}$; $C \in \{0.1, 1\}$
Random forest	300 trees; depth $\in \{\text{none}, 8, 16\}$; min. leaf $\in \{5, 20\}$; features $\in \{\sqrt{p}, 0.5\}$
Gradient boosting	learning rate $\in \{0.05, 0.1\}$; leaves $\in \{15, 31\}$; L2 $\in \{0, 1\}$

TABLE XIX
IMBALANCE STRATEGIES, LOGISTIC REGRESSION, AFFECTED, MAIN SET (MEAN OVER HELD-OUT REACHES; THRESHOLD 0.5 UNLESS STATED).

Strategy	AUC	Bal. acc.	F1	Brier
None	**0.971**	0.879	0.776	0.089
None, tuned threshold	**0.971**	0.904	0.748	0.089
Class weights	0.970	**0.910**	**0.777**	**0.078**
SMOTE	0.970	0.908	0.774	0.083
ISMOTE	0.939	0.856	0.667	0.140

class weights, SMOTE and a tuned threshold give almost the same balanced accuracy (0.90–0.91). ISMOTE lowers every measure for logistic regression (balanced accuracy 0.856, Brier 0.140). For random forest it matches SMOTE on affected (0.923) and is better on destroyed (AUC 0.938 vs. 0.924), but still below class-weighted logistic regression (0.964).

APPENDIX C
FEATURE GROUPS (E4)

Table XX adds the feature groups one at a time, with a fixed logistic regression and a random forest with 400 trees and minimum leaf 10 (E4 was run before the final settings were fixed, so this is not the forest of Table VIII), so that the gain of each group can be read directly. Local terrain and valley geometry each add about 0.01 to the affected AUC of logistic regression (Section V-F), and the same two steps raise destroyed from 0.942 to 0.964. The attenuation terms helped the destroyed targets most but hurt the near-channel subset ($0.856 \rightarrow 0.813$), because along-river distance is extrapolated for every held-out reach and, near the channel, mostly separates reaches with different damage shares. All features together gave no consistent gain: for logistic regression the full set was within 0.002 of the best subset on affected and destroyed but lower elsewhere; the random forest gained 0.012 on the near-channel subset ($0.826 \rightarrow 0.838$). The near-channel gains in this table come from models trained within the subset and do not survive when the fixed model is trained on all buildings (Section V-C).

APPENDIX D
INPUT SENSITIVITY (E8)

With a reduced, four-feature logistic regression (HAND, distance, slope, relief), results change little with the channel definition, the elevation model and the source of the negatives (Table XXI). Measuring HAND and distance to all rivers draining at least 200, 500 or 1,000 km^2 instead of to the traced flood path changes the mean affected AUC by at most 0.001, and FABDEM changes it by 0.002, although FABDEM lowers HAND by 1.5 m at the median and by more than 5 m for 16% of buildings. Keeping only OpenStreetMap footprints as negatives, which removes the machine-detected footprints that make the lower reach larger in our table than in the Copernicus inventory, changes affected by 0.001 and destroyed by 0.011. With the fixed ten-feature model, the OpenStreetMap-only table (11,580 buildings instead of 17,503; lower-reach prevalence 18% instead of 9%) gives a mean affected AUC of 0.972 (0.970 in the main table) and destroyed 0.956 (0.964); the main model scored on the same reduced table gives the same values, so the small change in destroyed comes from the smaller sample; training on it makes no difference.

APPENDIX E
DAMAGE REGIMES (E9)

K-means on the 3,360 affected buildings gave two weak clusters (silhouette 0.27 on a random subsample of 2,500 buildings, on which Ward clustering was also run; adjusted Rand index 0.73). One is an "open valley" regime (2,602 buildings, median HAND 18 m, valley width about 257 m, mostly middle and lower reaches); the other a "steep gorge" regime (758 buildings, median HAND 39 m, channel slope 2.5%, 82% upper reach). The share destroyed is similar (74% and 78%), so the clusters describe the setting of damage, and severity hardly differs between them (Fig. 8). The silhouette falls beyond two clusters (0.19 at $k = 3$, 0.15 at $k = 5$) and does not recover at $k = 6$ (0.153), and the Ward dendrogram splits first into the same two groups, so the structure among affected buildings is a single contrast between the gorge and the open valley.

APPENDIX F
COEFFICIENTS, PERMUTATION IMPORTANCE AND PARTIAL DEPENDENCE (E9)

a) Permutation importance on unseen reaches: The reach means are given in Section V-M; in single reaches the largest secondary drops are 0.036 and 0.027 (distance, lower and middle), 0.022 (slope, lower) and 0.012 (channel slope, upper). In the random forest with the attenuation set, the HAND $\times \log(d + 10)$ interaction carries most of the importance (0.18 vs. 0.04 for HAND), which is the attenuation curve of Section V-G expressed as a single feature.

TABLE XX

FEATURE GROUPS ADDED IN TURN: MEAN HELD-OUT ROC-AUC OVER THE THREE REACHES, LOGISTIC REGRESSION (LR, $C = 1$, CLASS WEIGHTS) AND RANDOM FOREST (RF, 400 TREES, MIN. LEAF 10). CORE: HAND AND DISTANCE; LOCAL: SLOPE, RELIEF, CURVATURES, WETNESS INDEX; GEOMETRY: VALLEY WIDTH, CHANNEL SLOPE, BEND; ATTENUATION: $\log(d + 10)$ AND ITS PRODUCT WITH HAND, USED IN PLACE OF GEOMETRY. N-CH.: NEAR-CHANNEL SUBSET, WITH THE MODELS TRAINED AND TESTED WITHIN THE SUBSET (MIDDLE AND LOWER REACHES).

	Affected		**Destroyed**		**Aff., n-ch.**		**D | A**	
Features	LR	RF	LR	RF	LR	RF	LR	RF
Core	0.954	0.924	0.942	0.909	0.832	0.774	0.748	0.630
+ local	0.964	0.928	0.951	0.924	**0.862**	0.814	0.759	0.750
+ geometry (main)	**0.970**	0.920	0.964	0.940	0.856	0.822	0.788	0.732
+ attenuation (replaces geometry)	0.968	**0.963**	0.971	0.962	0.813	0.826	**0.833**	**0.806**
All (geometry + attenuation)	0.968	0.961	**0.972**	**0.964**	0.824	**0.838**	0.820	0.788

TABLE XXI

INPUT SENSITIVITY OF A REDUCED, FOUR-FEATURE LOGISTIC REGRESSION (HAND, DISTANCE, SLOPE, RELIEF): MEAN HELD-OUT ROC-AUC OVER THE THREE REACHES; n = 17,503 EXCEPT THE OPENSTREETMAP-ONLY ROW (n = 11,580).

Elevation model	**Channel**	**Affected**	**Destroyed**
GLO-30	flood path (main)	0.963	0.947
GLO-30	rivers $\geq$ 200 km^2	0.963	0.949
GLO-30	rivers $\geq$ 500 km^2	0.963	0.949
GLO-30	rivers $\geq$ 1,000 km^2	0.963	0.947
FABDEM	flood path	0.961	0.946
FABDEM	rivers $\geq$ 500 km^2	0.962	0.948
GLO-30	flood path, OSM neg.[a]	0.964	0.936

[a]Only OpenStreetMap footprints kept as negatives.

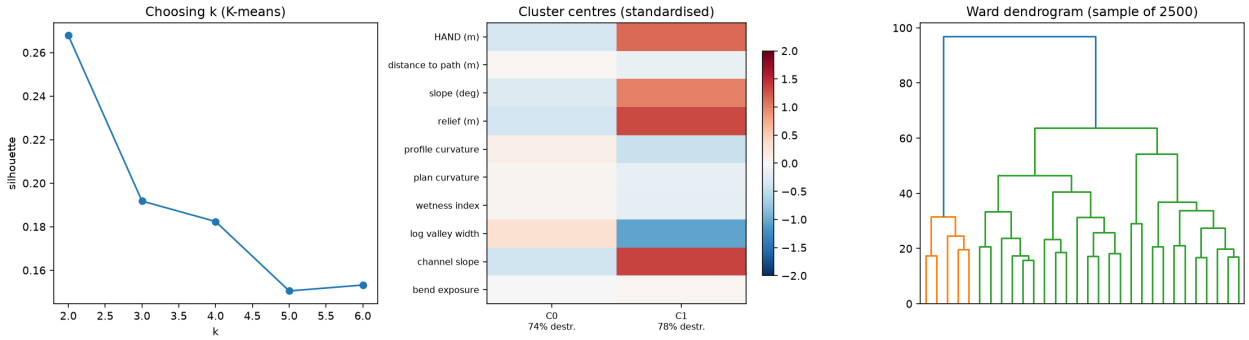


Fig. 8. Damage regimes among the 3,360 affected buildings. Left: silhouette by number of K-means clusters. Centre: standardised cluster centres. Right: Ward dendrogram on a sample of 2,500 buildings; its first split reproduces the two K-means clusters (adjusted Rand index 0.73).

b) Partial dependence and rules: Both models agree on the direction of every effect (Fig. 9). The random forest, however, becomes flat beyond about 40 m of HAND and 350 m of distance, while logistic regression keeps falling. This is the extrapolation behaviour seen in the area-of-applicability analysis. The depth-3 tree gives two readable rules for *affected*: (1) HAND $\leq$ 31 m and either within 316 m of the flood path or HAND $\leq$ 20 m; (2) HAND $>$ 31 m only where the channel slope exceeds 1.3% and the building is within 241 m of the path. The second rule is the gorge: there the flood reached higher.

APPENDIX G
ERRORS AND LAND COVER (E9)

a) Where the errors are: The errors follow the calibration result (Fig. 10, whose panels give the confusion counts per reach). In the lower reach, the threshold learned upstream predicts too much: 1,279 false positives against 10 false negatives. In the upper reach it predicts too little: 91 misses against 15 false positives, clustered in a few settlements of the gorge. The middle reach is in between (521 false positives, 143 false negatives). False positives line the river banks of the lower reach in continuous strips: the model ranks correctly along the valley but places the cut too high for the weaker flood.

b) Canopy and elevation-model error: Forest cover around each building was taken from the NLCMS 2022 [23] (class *forest*; share within a 5 $\times$ 5 window of 30 m pixels, about 75 m around the building). Only 1,594 buildings (9%) have at least 30% forest nearby, mostly on the valley sides. Canopy does raise HAND in the surface model: relative to FABDEM, GLO-30 places buildings 1.1 m higher where there is less than 10% forest and 4.7 m higher where there is more than 60% (medians). If this bias drove the errors, affected buildings near forest would be missed more often. They are not. Within strata of reach, HAND and distance, near-forest

TABLE XXII
STANDARDISED COEFFICIENTS OF THE CLASS-WEIGHTED LOGISTIC REGRESSION FITTED ON ALL BUILDINGS, WITH 95% INTERVALS FROM 300 BLOCK-BOOTSTRAP RESAMPLES (MAIN SET FOR AFFECTED AND DESTROYED; ATTENUATION SET FOR DESTROYED | AFFECTED, D | A). BOLD: INTERVAL EXCLUDES ZERO. –: FEATURE NOT IN THAT SET.

Feature	Affected	Destroyed	D | A
HAND	−6.12 (−8.59, −4.73)	−7.23 (−9.83, −5.69)	−3.82 (−5.79, −1.19)
Distance	−1.94 (−2.86, −0.87)	−1.85 (−2.94, −0.84)	−0.20 (−0.49, +0.08)
Slope	+0.66 (+0.42, +1.04)	+0.35 (+0.09, +0.74)	−0.13 (−0.65, +0.36)
Relief	+0.50 (+0.17, +0.95)	+0.35 (+0.01, +0.81)	+0.23 (−0.26, +0.79)
Profile curvature	−0.06 (−0.31, +0.39)	−0.20 (−0.54, +0.20)	−0.24 (−0.62, +0.09)
Plan curvature	+0.31 (+0.15, +0.48)	+0.33 (+0.17, +0.63)	+0.17 (−0.02, +0.50)
Wetness index	+0.11 (−0.03, +0.28)	+0.03 (−0.16, +0.22)	−0.19 (−0.43, +0.09)
Valley width (log)	−0.36 (−0.79, +0.12)	−0.35 (−0.69, +0.20)	–
Channel slope	+1.24 (+0.66, +1.89)	+1.47 (+0.92, +2.27)	–
Bend exposure	+0.06 (−0.30, +0.41)	+0.25 (+0.01, +0.56)	–
$\log(d+10)$	–	–	−3.11 (−4.17, −1.75)
HAND × $\log(d+10)$	–	–	+0.95 (−0.89, +2.68)

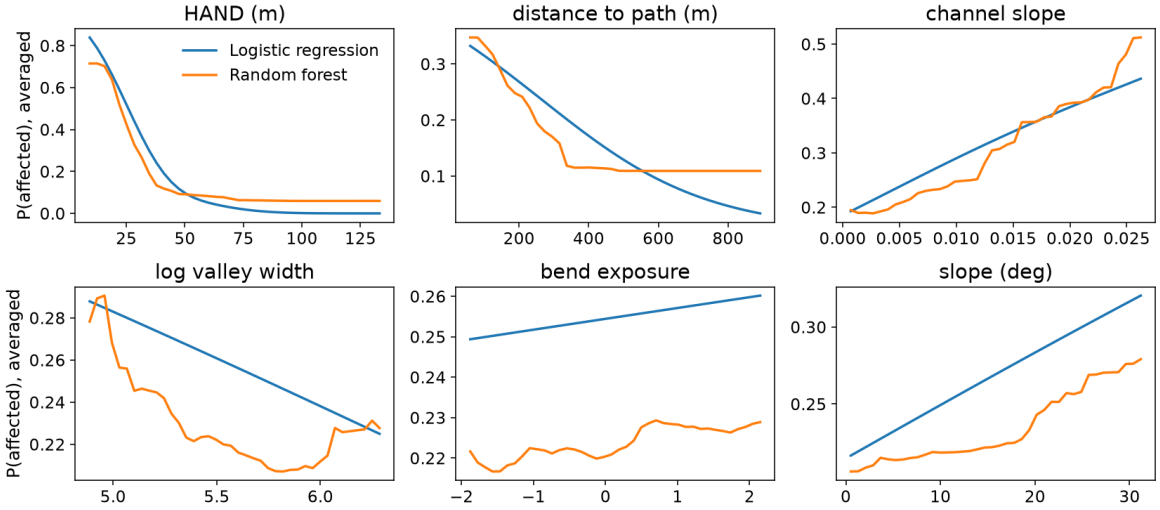


Fig. 9. Partial dependence of the probability of being affected on six terrain features, for class-weighted logistic regression and random forest fitted on all buildings (2nd–98th percentile range of each feature).

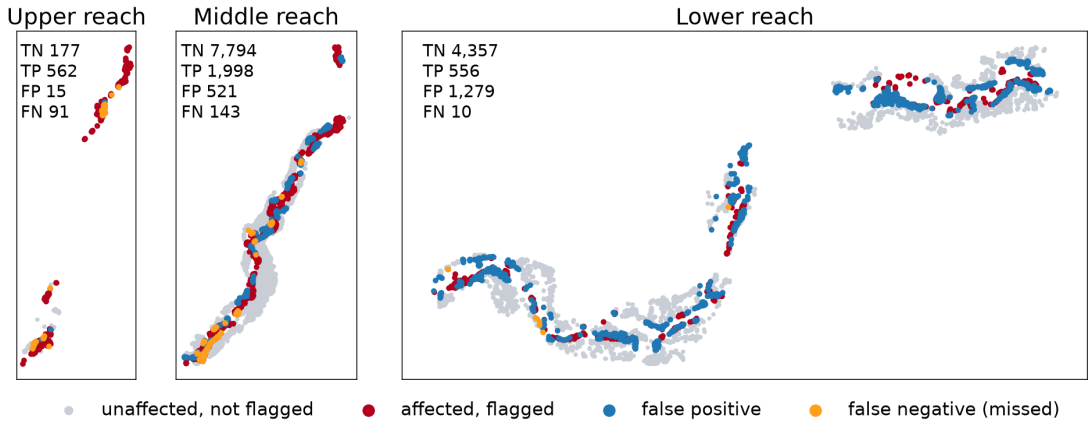


Fig. 10. Held-out errors of logistic regression (affected, main set, tuned threshold). Each panel shows one reach scored by a model trained on the other two. TP/TN: correct; FP: predicted affected but not; FN: affected but missed. Recall and precision at the tuned threshold: 0.86 and 0.97 (upper), 0.93 and 0.79 (middle), 0.98 and 0.30 (lower).

buildings are *less* likely to be missed (Mantel–Haenszel odds ratio 0.22, 95% Robins–Breslow–Greenland interval 0.07–0.74, $p = 0.01$) and more likely to be false positives (odds ratio 2.0, 1.4–3.0, $p < 10^{-4}$). The model scores them higher because they stand on steeper ground with more relief (median slope 10.5° vs. 3.8° in the middle reach), which it associates with

damage. ESA WorldCover 2021 [24] (10 m, class *tree cover*) flags more than four times as many buildings as near trees but gives odds ratios in the same direction (0.53, 0.35–0.80, $p = 0.003$, and 2.2, 1.9–2.7, $p < 10^{-17}$). The test therefore does not support canopy bias in the elevation model as the driver of the errors. The over-prediction near forest is consistent with the slope and relief effects.

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